ESTABLISHING EXTERNAL CRITERIA FOR UPDATING A META-ANALYSIS

UNIVERSIDAD COMPLUTENSE MADRID

Universidad Autónoma de Madrid

Santiago Arranz-Orlandi [1] and Juan Botella [2]

[1] Complutense University of Madrid | [2] Autonomous University of Madrid

INTRODUCTION

The continuous growth of the body of primary studies can cause **properly conducted meta-analyses (MA) to become outdated**.

The question of when it is justifiable to publish an update to a MA raises the need to establish **criteria that allow for the decision to update it**.

The existence of criteria independent of the meta-analysis's statistical significance can **guide a journal's editorial board** regarding the publication of MA updates.

Our objective is to develop **two** independent and complementary **external criteria** based on increases in (*a*) the **power of the test** and (*b*) the **precision of the estimate of the mean effect size**, understood as the width of the confidence interval. We also aim to develop an R *script* that conveniently allows for the estimation of the additional studies needed to meet these a priori established criteria.

DEVELOPMENT OF THE CRITERIA

Power

The power of a test is the probability of rejecting the null hypothesis (H_0) when it is false. It is determined by the **significance level** (α), the alternative **parametric effect size (TE)** for which the power is to be determined (θ_1), and the **standard error of** the test **statistic** (σ). For a right-tailed one-sided test, it is calculated—taking $TE_{\bullet(c)}$ as the smallest effect size combined with which the null hypothesis test would be significant—as follows (Hedges & Pigott, 2001):

$$pot = 1 - \Phi\left(\frac{TE_{\bullet(c)} - \theta_1}{\hat{\sigma}_{\theta_\bullet}}\right)$$

If we set an a priori power value, we can **estimate the number of studies** required to achieve that value by first solving for the standard error in the above formula:

$$\hat{\sigma}_{\theta_\bullet} = \frac{\theta_1}{z_{1-\alpha} - z_\beta}$$

Where $z_{1-\alpha}$ corresponds to the value of the standard normal distribution with a cumulative probability of $1-\alpha$, and z_β corresponds to the value of the same distribution with a cumulative probability equal to the probability of committing a Type II error ($\beta = 1 - pot$).

Range

The meta-analytic estimate of the combined effect size provides both a point estimate and a range of plausible values within which the mean of the parametric effect sizes might lie. This range is known as **the confidence interval**, and its width is a measure of the precision of the estimate. Assuming an asymptotic normal distribution, the width can be expressed as twice the standard error multiplied by the value of the standard normal distribution corresponding to the (1 – α/2) percentile:

$$a = 2\cdot\left(\left|z_{1-\alpha/2}\right|\cdot\hat{\sigma}_\theta\right)$$

If we set an a priori width value, we can **estimate the number of studies** needed to achieve that value by first isolating the standard error from the previous formula:

$$\hat{\sigma}_{(k+j)} = \frac{a'}{2\cdot\left|z_{1-\alpha/2}\right|}$$

Study Estimation

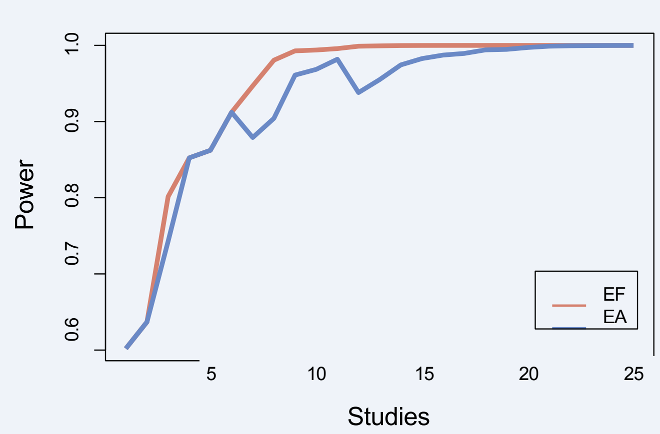
The square of the standard error (variance) is equal to the inverse of the sum of the weights, both those from the original meta-analysis (*k*) and those from the studies to be added (*j*): $W_{(k+j)} = 1/\hat{\sigma}^2_{(k+j)}$. The **weight that must be added to achieve the** a priori **determined power or effect size** is obtained as: $W_j = W_{(k+j)} - W_k$.

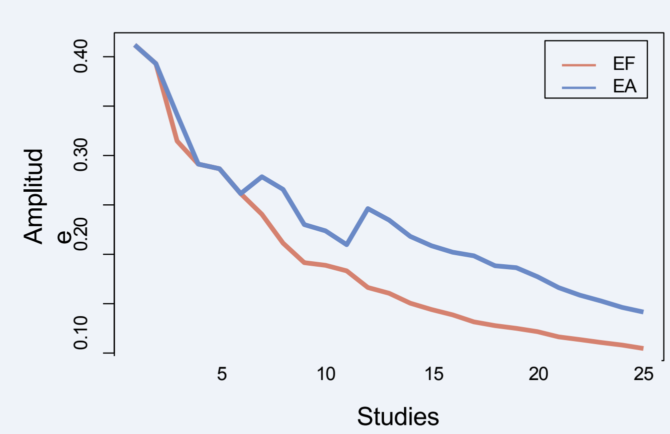
This weight can be expressed in various ways. We will primarily treat it as **several studies of sizes similar** to those already included in the meta-analysis. To calculate a reasonable number of studies needed to achieve $W_{(j)}$, one can assume that the sizes of the studies already conducted are representative of studies in that field and, therefore, future studies can be expected to be of a similar size. In summary, it is assumed that **each study to be added has a weight equivalent to the average weight** ($\bar{w}$) of the original studies: $\bar{w} = W_k / k$. Therefore, the number of studies to be added is approximately $j \approx W_j / \bar{w}$.

Fixed-effect and random-effects models

Under a fixed-effects model (all estimates are of a single parametric treatment effect), aggregating studies into a meta-analysis will always result in a reduction in the standard error, since the weight of a study depends only on the sampling variance: $w_i^{EF} = 1/\hat{\sigma}^2_{m(i)}$. If a **random-effects model** is used (each estimate is of a parametric treatment effect within a population of effects), the weight of the studies depends on both the sampling variance and the between-study variance (τ^2): $w_i^{EA} = 1/\left(\hat{\sigma}^2_{m(i)} + \hat{\tau}^2\right)$. This variance is estimated based on the variability of the studies included in the meta-analysis; therefore, its estimate may vary if studies are included with a treatment effect and/or variance that differs significantly from that of the other studies. **Re-estimating the between-study variance may result in a reduction in the increase in power or in the reduction in effect size**.

Simulation of the increase in power and the decrease in effect size when adding studies with highly variable data





For our calculations, we assume that the data from the original meta-analyses are a representative sample of the study population, so the estimate of τ^2 should be reliable. Furthermore, our estimation procedure assumes that the studies to be added are similar to those already included in the original meta-analysis; therefore, **in the long run, the between-study variance should remain stable**. If we were to re-estimate τ^2 with each "similar" study added, it would decrease drastically and artificially, biasing the estimate.

DISCUSSION AND FUTURE DIRECTIONS

This study serves as a first step toward establishing external criteria to guide the updating of a meta-analysis. **The value of the criteria should be guided by the substantive field** of the meta-analysis.

These calculations should be extended to the testing of moderator effects, which typically have much lower statistical power (Valentine et al., 2010).

The potential influence of publication bias on the validity and realism of these criteria should also be investigated. Finally, it would be advisable to continue developing the R code, adding more features—such as the calculation of the standard error using the method of Hartung and Knapp (2001), and making it available to the public as a package.

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